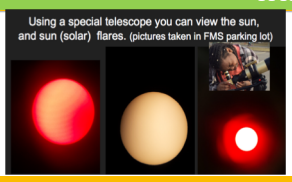
Evolution of a Conceptual Framework for Studying Science Voice Development

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University of Minnesota Driven to Discover℠

Initial Proposal: Four Dimensions and Related Manifestations



Participant science voice development was evaluated using:

- social network analysis of a secure online forum.
- written responses to prompts in digital and paper notebooks.
- verbal responses to prompts on Flipgrid.
- participant created astronomical images.
- research project presentations outside the classroom.
- badges earned by completing Skynet explorations.
- semistructured interviews.

Active Participation
- increased knowledge
- immersive experience
- exposure to multiple models
- appropriation of scientific language
- learning integrated in situational practice

Interaction with a Learning Community
- exhibit a sense of belonging
- practice using scientific language
- productive interaction with
  - peers
  - professionals

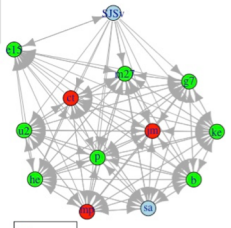
Science Voice

Express a Science Identity
- interact with a community of scientific practice
- establish and make progress toward scientific goals
- bridge current cultural identity(ies) with components of science culture
- exhibit
  - confidence / authority
  - agency
  - optimism
  - expanded interest in science content

Demonstrate Self Efficacy
- display unique presence
- demonstrate individual expression
- perform authentic scientific research
- express personally meaningful results
- exchange information in multiple modes
- communicate to a community of scientific practice for
  - examination
  - learning
  - critique

Participants exhibited belonging, interacted with a communities of practice and exchanged information through project collaboration and forum exchanges.

SJS "Scavenger Hunt" Forum Discussion

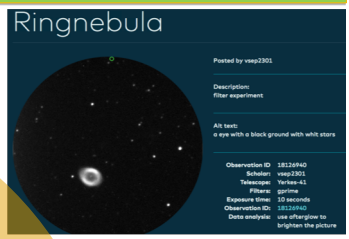


size	density	reciprocity	diameter	network size
13	0.865	0.84	2	139

Modified Proposal: Three Dimensions and Related Manifestations



Participants demonstrated creativity and unique cultural expressiveness as they created and shared personally meaningful astronomical images.



Interviewer: ***"Tell me about your experience with our after-school astronomy program."***
Quin: ***"It was fun. When I first came and I wasn't sure though but I stayed. We have to learn about a lot of things like Messier 1 and how the Crab Nebula looks. It's cool and fun."***

Express a Science Identity
- exhibit expanded interest
- bridge cultural identities
- establish scientific goals
- self reflection

Active Participation
- situational practice
- practice scientific skills
- perform scientific research

Interact With A Science Community

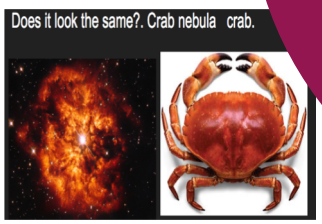
Science Voice

Scientific Language Appropriation

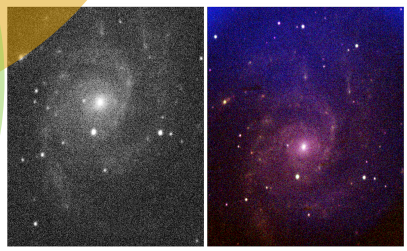
Communicate in Multiple Modes

Demonstrate Self Efficacy
- share personally meaningful results
- exhibit a sense of belonging
- increase science knowledge
- accomplish scientific goals

Interviewer: ***"What do you like about it?"***
Aran: ***"Everything. I'm learning a lot like I never knew, like it takes light years, I didn't even know what light years were."***



Sample Participant Images



Badges earned by participants in our pilot study.

Activity	First Light	Exposure Time	Scavenger Hunt	Investigate Filters	Make a Color Image	Create it Yourself
Student						
Vivian	X	X	X	X	X	
Jay	X	X	X	X		
Don	X	X	X			
Quin						
Aran	X	X				

Flipgrid Prompt: ***"Tell me about the first picture you took."*** **Vivian:** ***"The first picture I took was a nebula. It kind of looked like teeth. I used the clear filter and the telescope I used Yerkes 41."***

Implications

- **Social justice implications of science voice research:**
  - **reduce alienation students may feel towards science.**
  - **increase agency in science.**
  - **empower students to cross science cultural borders.**
- **Encourages students to become producers of knowledge instead of just consumers.**

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Related Paper Available Here

Preliminary Data From Spring 2017 Pilot Case Study of an Informal Astronomy Project

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